

AI-POWERED JOB RECRUITMENT & CANDIDATE MANAGEMENT SYSTEM

A Salesforce CRM & Agentforce AI Implementation

Project Documentation & Technical Report

Prepared By	Gampala Tharun Reddy
Email	tharunreddy_gampala@srmap.edu.in
Application Number	AP24110010090
Platform	Salesforce (Declarative / Admin + Agentforce AI)
Document Type	Project Report & Implementation Documentation

Table of Contents

1. Project Overview

2. Data Model

Phase 1: Requirement Analysis and Planning

Phase 2: Salesforce Development – Backend & Configuration

Milestone 1: Salesforce Account

Milestone 2: Objects Creation

Milestone 3: Tabs

Milestone 4: Fields & Relationships

Milestone 5: Validation Rules

Milestone 6: Approval Process

Milestone 7: Email Templates

Milestone 8: Email Alerts

Milestone 9: Declarative Automation (Flows)

Milestone 10: Agentforce AI

Phase 3: UI/UX Development & Customization

Milestone 11: The Lightning App

Milestone 12: Editing of Page Layouts

Milestone 13: Dynamic Forms

Milestone 14: Users

Phase 4: Data Migration, Testing & Security

Milestone 15: Duplicate and Matching Rules

Milestone 16: Profiles

Milestone 17: Roles & Role Hierarchy

Phase 5: Deployment, Documentation & Maintenance

6. Conclusion

7. Project Author

Tip: use your PDF reader's Bookmarks / Outline panel for one-click navigation to any section.

1. Project Overview

The AI-Powered Job Recruitment & Candidate Management System is a centralized Salesforce application built to streamline and modernize the end-to-end hiring lifecycle using declarative automation and Agentforce artificial intelligence. The system was designed and implemented entirely using Salesforce Admin tools (Setup, Flow Builder, Prompt Builder, and Agentforce) without custom Apex development, demonstrating that a complete, production-ready recruitment platform can be delivered through configuration and AI-native automation.

Modern hiring teams commonly struggle with duplicate candidate records, lost applications, delayed interviews, inconsistent hiring decisions, and poor visibility into recruitment metrics because they rely on spreadsheets, emails, and disconnected tools. This project addresses these problems by providing a single, secure, and intelligent CRM-based hiring platform.

1.1 Why Salesforce

- Secure, scalable cloud infrastructure
- Built-in declarative automation (Flows, Approval Processes, Validation Rules)
- Native AI capability through Agentforce (Prompt Builder and AI Agents)
- Real-time reporting and dashboards
- Robust, layered security model (Profiles, Roles, Sharing Rules, Field-Level Security)

1.2 Objectives

- Implement core Salesforce CRM fundamentals for a hiring use case
- Design a scalable recruitment data model (Candidate, Job Opening, Candidate Application)
- Apply security at the org, object, field, and record level
- Automate recruitment workflows using Record-Triggered, Scheduled, and Screen Flows
- Use Agentforce AI for candidate summarization and hiring-risk analysis
- Enable approval-driven decision-making for high-value job offers

1.3 Recruitment Lifecycle Supported

Candidate → Screening → Interview → Offer → Hiring

2. Data Model

Three core entities drive the recruitment process: the standard Contact object (representing candidates), and two custom objects, Job Opening and Candidate Application, connected through master-detail relationships.

2.1 Contact (Candidate)

- Candidate ID – Auto Number (C-{0000})
- Experience Level – Picklist (Fresher, Mid-Level, Senior)
- Application Status – Picklist (New, Screening, Interview, Offer, Hired)

2.2 Job Opening (Job_Opening__c)

- Department – Picklist (IT, HR, Finance, Sales)
- Salary Range – Currency
- Vacancies – Number
- Job Status – Picklist (Open, Closed, On Hold)

2.3 Candidate Application (Candidate_Application__c)

- Candidate – Master-Detail (Contact)
- Job Opening – Master-Detail (Job Opening)
- Application Date – Date
- AI Candidate Score – Number
- Offer Amount – Currency
- Application Status – Picklist (Active, Inactive)
- Approval Status – Picklist (Pending, Approved, Rejected)

Phase 1: Requirement Analysis and Planning

Complexity: Medium | Duration: 1h 0m

Understanding Business Requirements

Modern organizations receive hundreds of candidate applications daily. Most hiring teams rely on spreadsheets, emails, and disconnected tools, which leads to duplicate candidate records, lost applications, delayed interviews, inconsistent hiring decisions, lack of transparency, and poor tracking of recruitment metrics. The business requires a centralized CRM-based hiring system to track all candidates, standardize screening, automate hiring workflows, provide AI-driven evaluation, improve recruiter productivity, and support managerial approvals.

Defining Project Scope and Objectives

The project scope covers building a Salesforce recruitment system using Admin tools and Agentforce AI without custom Apex development, supporting the full recruitment lifecycle from candidate application through screening, interview, offer, and hiring.

Phase 2: Salesforce Development – Backend & Configuration

Complexity: Medium | Duration: 8h 54m

This phase covers the core declarative build of the system: Salesforce org setup, custom objects, tabs, fields and relationships, validation rules, the approval process, email templates and alerts, automation flows, and the configuration of Agentforce AI.

Milestone 1: Salesforce Account

Assigned To: Bhavya Inturi | Duration: 20m

A Developer Edition Salesforce org was created to serve as the foundation of the recruitment system. This org hosts all custom objects, automation, security configuration, and Agentforce AI components used throughout the project.

Screenshot Reference

Screenshots/01_Account_Setup/M01_Salesforce_Home_Page.png
Screenshots/01_Account_Setup/M01_Setup_Page.png

Milestone 2: Objects Creation

Assigned To: Hariteja Dasari | Duration: 30m

Two custom objects were created to represent hiring data: Job Opening (Record Name: Job Title, Text) and Candidate Application (Record Name: Application ID, Auto Number format CA-{0000}). Both objects were configured with Allow Reports, Allow Search, and Track Field History enabled.

Screenshot Reference

Screenshots/02_Objects/M02_Job_Opening_Object_Form.png
Screenshots/02_Objects/M02_Job_Opening_Object_Created.png
Screenshots/02_Objects/M02_Candidate_Application_Object_Form.png
Screenshots/02_Objects/M02_Candidate_Application_Object_Created.png

Milestone 3: Tabs

Assigned To: Hariteja Dasari | Duration: 20m

Custom tabs were created for the Job Opening and Candidate Application objects so that users can navigate to and manage recruitment records directly from the Salesforce navigation bar.

Screenshot Reference

Screenshots/03_Tabs/M03_Job_Opening_Tab_Form.png.png
Screenshots/03_Tabs/M03_Job_Opening_Tab_Created.png
Screenshots/03_Tabs/M03_Candidate_Application_Tab_Form.png

Screenshots/03_Tabs/M03_Candidate_Application_Tab_Created.png

Milestone 4: Fields & Relationships

Assigned To: Pranitha Chinnapureddy | **Duration:** 39m

Fields were added across the Contact, Job Opening, and Candidate Application objects, including picklists, currency, number, date, and master-detail relationship fields, to fully capture candidate, job, and application data. Master-detail relationships link Candidate Application to both Contact and Job Opening.

Screenshot Reference

Screenshots/04_Fields_Relationships/M04.Contact_Fields/Candidate_ID.png
Screenshots/04_Fields_Relationships/M04.Contact_Fields/Experience_Level.png
Screenshots/04_Fields_Relationships/M04.Contact_Fields/Application_status.png
Screenshots/04_Fields_Relationships/M04.JobOpening_Fields/Department.png
Screenshots/04_Fields_Relationships/M04.JobOpening_Fields/Salary_Range.png
Screenshots/04_Fields_Relationships/M04.JobOpening_Fields/Vacancies.png
Screenshots/04_Fields_Relationships/M04.JobOpening_Fields/Job_Status.png
Screenshots/04_Fields_Relationships/M04.CandidateApplicationFields/Candidate.png
Screenshots/04_Fields_Relationships/M04.CandidateApplicationFields/
Job_Opening.png
Screenshots/04_Fields_Relationships/M04.CandidateApplicationFields/
Applicatio_Date.png
Screenshots/04_Fields_Relationships/M04.CandidateApplicationFields/
AI_Candidate_Score.png
Screenshots/04_Fields_Relationships/M04.CandidateApplicationFields/
Offer_Amount.png
Screenshots/04_Fields_Relationships/M04.CandidateApplicationFields/
Application_Status.png
Screenshots/04_Fields_Relationships/M04.CandidateApplicationFields/
Approval_Status.png

Milestone 5: Validation Rules

Assigned To: Gampala Tharun Reddy | **Duration:** 45m

Three validation rules enforce data integrity: Offer Amount Validation (Offer_Amount__c <= 0 blocked, ensuring the offer amount is always greater than zero), Job Vacancies Validation (Vacancies__c < 0 blocked, preventing negative vacancy counts), and AI Candidate Score Validation (AI_Candidate_Score__c > 100 blocked, capping the AI score at 100).

Screenshot Reference

Screenshots/05_Validation_Rules/M05_Offer_Amount_Validation.png
Screenshots/05_Validation_Rules/M05_Job_Vacancies_Validation.png
Screenshots/05_Validation_Rules/M05_AI_Candidate_Score_Validation.png

Milestone 6: Approval Process

Assigned To: Hariteja Dasari | **Duration:** 40m

An Approval Process (“High Salary Candidate Offer Approval”) was configured on the Candidate Application object. It is triggered when the Offer Amount exceeds 1,000,000 and the Application Status equals “Offer Submitted.” The record owner’s manager is automatically assigned as the approver, only Administrators can edit records while a request is pending, and Approval Status is automatically updated to Approved or Rejected via field updates upon the final decision. The record owner (recruiter) is set as the initial submitter.

Screenshot Reference

Screenshots/06_Approval_Process/M06_Final_approval_action.png

Screenshots/06_Approval_Process/M06_Approval_processes_activated.png

Milestone 7: Email Templates

Assigned To: Gampala Tharun Reddy | **Duration:** 30m

Three Classic Email Templates were created: High Priority Candidate Application Notification (alerts the recruiter when AI identifies a high-priority candidate), High Salary Candidate Offer Approval Notification (notifies the hiring manager that a high-salary offer needs review), and Candidate Application Approval Status Notification (informs the recruiter of the final approval outcome). All templates dynamically merge candidate, job opening, offer, and AI score fields.

Screenshot Reference

Screenshots/07_Email_Templates/M07_Email_Tempelate.png

Milestone 8: Email Alerts

Assigned To: Bhanu Sri Inturi | **Duration:** 30m

Email Alerts were built on the Candidate Application object to connect the email templates to automation: a High Priority Candidate Application Email Alert sent to the application owner, a High Salary Candidate Offer Approval Email Alert sent to the record owner’s manager, and a Candidate Application Approval Status Email Alert sent back to the application owner once a decision is made.

Screenshot Reference

Screenshots/08_Email_Alerts/M08_Email_alert.png

Milestone 9: Declarative Automation (Flows)

Assigned To: Gampala Tharun Reddy | **Duration:** 2h 0m

Four flows automate the recruitment process. A Record-Triggered Flow sends the High Priority Candidate Application Notification whenever AI_Candidate_Score__c ≥ 85. A second Record-Triggered Flow automatically

submits the record for approval whenever Offer_Amount__c > 1,000,000. A Scheduled-Triggered Flow runs daily to identify applications pending approval for more than three days and reminds the recruiter. A Screen Flow (“Candidate Application Screen Flow”) walks recruiters through a guided, multi-screen wizard — capturing candidate, job opening, application date, offer amount, and experience level — before creating the Candidate Application record and calculating the final offer amount.

Screenshot Reference

Screenshots/09_Flows/M09_High_Priority_Alert.png
Screenshots/09_Flows/M09_Salary_Approval_Flow.png
Screenshots/09_Flows/M09_Pending_Reminder_Flow.png
Screenshots/09_Flows/M09_Offer_Calculation_Flow.png

Milestone 10: Agentforce AI

Assigned To: Hariteja Dasari | **Duration:** 2h 40m

Agentforce and AI Trust settings (Data Masking, Zero Data Retention, Audit Logging) were enabled for the System Administrator and Hiring Manager profiles. Two Prompt Templates were built in Prompt Builder: a Candidate Application Summary Prompt (Summary type) that condenses each application into a readable overview, and a Candidate Hiring Risk Analysis Prompt (Flex type) that evaluates hiring probability, financial risk, and overall risk level (Low/Medium/High) with an approve-or-reject recommendation.

A Recruitment Assistant Agent was then created and configured with topics for Candidate Application Summary, Hiring Risk Analysis, and Offer Approval Guidance, and actions mapped to the two prompt templates plus a Recommend Approval Decision action. Its Data Library was connected — read-only, with field-level access controls — to Contact, Job_Opening__c, and Candidate_Application__c so the agent can answer recruitment questions directly from Salesforce data. The agent was tested on live records (“Summarize this candidate application,” “Is this candidate safe to hire?”, “What risks are involved in this salary offer?”) and deployed to the Lightning Experience for Recruiters and Hiring Managers.

Screenshot Reference

Screenshots/10_Agentforce/M10_Candidate_Application_Summary_Prompt_Activated.png
Screenshots/10_Agentforce/M10_Candidate_Hiring_Risk_Analysis_Prompt_Creation.png
Screenshots/10_Agentforce/M10_RecruitmentAssistant_Agent_Active.png
Screenshots/10_Agentforce/M10_Recruitment_Assistant_Agent_Active.png
Screenshots/10_Agentforce/M10_Agentforce_Test_Completion_Screenshot.png

Phase 3: UI/UX Development & Customization

Complexity: Medium | **Duration:** 2h 0m

This phase focuses on building an intuitive, secure, and productivity-driven interface so recruiters and hiring managers can view AI-driven candidate insights, prioritize high-value applications, evaluate risk recommendations, manage approval-driven offers, and track application status efficiently. A Lightning App, customized Page Layouts, Dynamic Forms, and dedicated user accounts were configured in this phase.

Milestone 11: The Lightning App

Assigned To: Gampala Tharun Reddy | **Duration:** 20m

A dedicated Lightning App, “AI-Powered Recruitment & Candidate Management System,” was created through App Manager. It includes the Candidate Applications, Candidates (Contact), Job Opening, Accounts, Reports, and Dashboards tabs, and was assigned to the System Administrator, Recruiter, and Hiring Manager profiles.

Screenshot Reference

Screenshots/11_Lightning_App/M11_Final_App_Created.png

Milestone 12: Editing of Page Layouts

Assigned To: Gampala Tharun Reddy | **Duration:** 50m

Page layouts were customized for Candidate Application, Job Opening, Account, Opportunity, and Contact to present recruitment-relevant fields — including AI Candidate Score, Offer Amount, and Approval Status — in a clear, prioritized arrangement for end users.

Screenshot Reference

Screenshots/12_Page_Layouts/M12_Candidate_Application_Layout.png

Screenshots/12_Page_Layouts/M12_Job_Opening_Layout.png

Screenshots/12_Page_Layouts/M12_Account_Layout.png

Screenshots/12_Page_Layouts/M12_Opportunity_Layout.png

Screenshots/12_Page_Layouts/M12_Contact_Layout.png

Milestone 13: Dynamic Forms

Assigned To: Bhavya Inturi | **Duration:** 30m

Dynamic Forms were enabled on the Candidate Application record page (and related objects) by upgrading the Details section, selecting the corresponding page layout, and activating the form as the org default for both desktop and phone.

Screenshot Reference

Screenshots/13_Dynamic_Forms/M13_Upgrade_Dynamic_Forms.png

Screenshots/13_Dynamic_Forms/M13_Dynamic_Forms_Activated.png

Screenshots/13_Dynamic_Forms/M13_Candidate_Application_Record.png

Milestone 14: Users

Assigned To: Hariteja Dasari | **Duration:** 20m

Dedicated user accounts were created for a Recruiter (Role: Recruiter, Profile: Recruiter Profile) and a Hiring Manager (Role: Hiring Manager, Profile: Hiring Manager Profile), both provisioned with the Salesforce user license, so that role-based access and the approval hierarchy could be fully validated.

Screenshot Reference*Screenshots/14_Users/M14_Recruiter_User_Creation.png**Screenshots/14_Users/M14_Hiring_Manager_User_Creation.png*

Phase 4: Data Migration, Testing & Security

Complexity: Medium | Duration: 1h 10m

This phase focused on migrating candidate, job opening, and application data into Salesforce, validating end-to-end AI-assisted recruitment workflows, and enforcing robust security controls, followed by performance tuning to support multiple concurrent users.

4.1 Data Migration

- Candidate data imported using the Data Import Wizard
- Job Opening and Candidate Application data imported using Data Loader
- Candidate Email and Application ID used as unique identifiers to prevent duplicates
- Relationships verified between Candidate, Candidate Application, and Job Opening
- Migrated data validated using Reports and List Views

4.2 Testing

- Unit Testing – AI candidate score logic, hiring risk analysis logic, approval status updates
- Integration Testing – Flows, Approval Processes, Email Alerts, and Agentforce prompt actions working together
- User Acceptance Testing (UAT) – conducted with Recruiter, Hiring Manager, and System Administrator sample users covering the full application-to-hiring journey

Milestone 15: Duplicate and Matching Rules

Assigned To: Pranitha Chinnapureddy | Duration: 20m

A custom Matching Rule (“Unique Candidate – Email & Phone”) uses exact matching on Email and Phone to identify duplicate Contact records. A corresponding Duplicate Rule (“Unique Candidate Email and Phone”) reports and alerts users on create and edit when a match is found, protecting the accuracy of AI candidate scoring, risk analysis, and reporting.

Screenshot Reference*Screenshots/15_Duplicate_Rules/M15_Matching_Rules_Page.png**Screenshots/15_Duplicate_Rules/M15_Matching_Rule_Activated.png*

Screenshots/15_Duplicate_Rules/M15_Duplicate_Rule_Activated.png

Milestone 16: Profiles

Assigned To: Gampala Tharun Reddy | **Duration:** 20m

Two custom profiles were cloned from Standard User. The Recruiter Profile grants Read/Create/Edit on Contact, Candidate Application, and Job Opening, read-only access to Reports and Dashboards, read-only visibility of the AI Candidate Score, no approval or override permissions, a 2-hour session timeout, and a 90-day password expiry policy. The Hiring Manager Profile grants full Read/Create/Edit/Delete access on the same objects plus Reports and Dashboards, the ability to approve/reject high-salary offers and override AI recommendations, and full visibility into AI Candidate Score, Risk Analysis, and Hiring Recommendations.

Screenshot Reference

Screenshots/16_Profiles/M16_Profiles_Page.png

Screenshots/16_Profiles/M16_Recruiter_Profile_Created.png

Screenshots/16_Profiles/M16_Recruiter_Profile_Permissions.png

Screenshots/16_Profiles/M16_Hiring_Manager_Profile_Created.png

Screenshots/16_Profiles/M16_Hiring_Manager_Profile_Permissions.png

Milestone 17: Roles & Role Hierarchy

Assigned To: Bhavya Inturi | **Duration:** 30m

A two-tier role hierarchy was established under the default CEO role: Hiring Manager at the top, with Recruiter reporting into Hiring Manager. This ensures recruiters see only their assigned candidates and applications while hiring managers retain visibility across their team's recruitment records, in line with the org-wide defaults and sharing rules defined for the system.

Screenshot Reference

Screenshots/17_Roles/M17_Final_Role_Hierarchy.png

4.3 Security Implementation

Role Hierarchy

- System Administrator – Full access
- Hiring Manager – Team visibility, approvals, AI insights
- Recruiter – Manage assigned candidates and applications

Field-Level Security

Sensitive fields — AI Candidate Score, Risk Level, Proposed Offer Amount, and Approval Status — are visible only to Hiring Managers and System Administrators and hidden from recruiters where required.

Record-Level Security

- Org-Wide Defaults: Candidate = Private, Candidate Application = Private, Job Opening = Public Read Only
- Sharing Rules: Recruiters access their assigned records; Hiring Managers have read/write access to team records

4.4 Performance Tuning & Optimization

Flow Optimization

- Entry conditions applied early to avoid unnecessary flow execution
- Bulk-safe automation with no updates inside loops
- Scheduled flows used efficiently for reminder logic

Page Layout & Security Enhancements

- Streamlined layouts for Candidate, Job Opening, and Candidate Application
- Formula fields used for offer calculations and risk indicators
- Field-level security enforced across flows and UI; Agentforce prompts restricted by object access

Phase 5: Deployment, Documentation & Maintenance

Complexity: Medium | Duration: 1h 28m

5.1 Deployment Strategy

The system was deployed following Salesforce best practices to ensure stability, data security, and minimal disruption to ongoing recruitment activities. Flows, Approval Processes, Email Alerts, Agentforce configurations, and custom metadata were validated in the sandbox, followed by final regression testing on candidate application creation, AI prioritization, high-salary offer approvals, and email notifications, before deploying custom objects, fields, flows, approval processes, email assets, Lightning components, and Agentforce configurations to production.

5.2 User Training & Enablement

Training was delivered to Recruiters, Hiring Managers, and System Administrators through live demonstrations, scenario-based walkthroughs, hands-on practice with sample recruitment data, and a Q&A/feedback session, covering topics from interpreting AI candidate scores and risk analysis to managing agents, prompt templates, and automation thresholds.

5.3 User Adoption & Change Management

AI and Agentforce components were embedded directly on the Home Page, the Candidate Application record page, and within the Lightning App to minimize the learning curve and encourage adoption of AI-assisted recruitment decisions.

5.4 Monitoring & Post-Deployment Support

- Tracked usage of AI candidate scoring and recommendations
- Monitored hiring conversion rates and approval turnaround times
- Reviewed performance of flows and scheduled automations
- Fine-tuned AI scoring thresholds based on real hiring data and improved UI clarity

5.5 Business Impact & Outcomes

- Faster hiring decisions and better prioritization of high-quality candidates
- Reduced manual evaluation effort through AI-assisted scoring
- Improved offer approval efficiency via automated routing
- Data-driven hiring decisions and increased recruitment team productivity

6. Conclusion

The AI-Powered Job Recruitment & Candidate Management System demonstrates that a complete, intelligent, and secure hiring platform can be delivered on Salesforce using purely declarative tools — custom objects, validation rules, approval processes, Flow Builder automation, and Agentforce AI — without writing custom Apex code. From candidate intake through AI-driven scoring, risk analysis, approval-based offer management, and role-based security, the system covers the full recruitment lifecycle end to end.

This documentation, together with the accompanying milestone-wise screenshots, serves as a complete implementation record and reference for future enhancements, onboarding, and maintenance of the system.

7. Project Author

Name	Gampala Tharun Reddy
Email	tharunreddy_gampala@srmmap.edu.in
Application Number	AP24110010090